

Compton production and detection of dark photons

TEXONO: vacuum calculation, medium correction, and published limits

ROOT/C++

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Abstract

The tree-level production cross section passes independent matrix, Ward-identity, total-integral and massless-limit checks. Recalculating Park’s Figure 1 does not reproduce the full published spectrum; the disagreement is displayed without fitting. For TEXONO we compare Park’s quoted bound, the curve extracted from Danilov’s PDF, an independent calculation with Park’s conventions, and a homogeneous-medium calculation. The medium result includes an explicit signal-selection efficiency. No exclusion is claimed at material resonance or outside the stated small-mixing domain.

1 What is derived, and what is taken from sources

We reuse and recheck the project’s v2 polynomial trace kernel. QED vertices, the electron propagator, spin completeness and two-body phase space are input identities. The compact trace, numerical convolution, inverse cross section and exclusion solver are implemented here. Gondolo–Raffelt (Park’s Reference 15) supplies an independent total-cross-section check; it is not required to generate the differential kernel. Park supplies the empirical source and experimental inputs. Danilov supplies the medium probabilities and selection correction.

2 Production amplitude and trace

$$\gamma(k) + e(p) \rightarrow X(q) + e(p'), \quad p^2 = p'^2 = m^2, \quad k^2 = 0, \quad q^2 = M^2, \quad g = \text{diag}(1, -1, -1, -1). \quad (1)$$

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi(A_\mu + \epsilon X_\mu), \quad (2)$$

$$\mathcal{M} = \epsilon e^2 \bar{u}(p') \left[\gamma^\nu \frac{\not{p} + \not{k} + m}{s - m^2} \gamma^\mu + \gamma^\mu \frac{\not{p}' - \not{q} + m}{u - m^2} \gamma^\nu \right] u(p) \varepsilon_\mu \xi_\nu^*. \quad (3)$$

The two electron-exchange diagrams are both required. On-shell Dirac equations give zero upon contraction with either external vector momentum. Therefore the massive polarization sum can be replaced by the metric in the squared amplitude; the longitudinal term vanishes by current conservation. This does not discard a physical polarization.

$$|\overline{\mathcal{M}}|^2 = \epsilon^2 e^4 F, \quad a = s - m^2, \quad b = u - m^2, \quad c = m^2, \quad d = M^2, \quad (4)$$

$$F = -2 \left(\frac{a}{b} + \frac{b}{a} \right) + 4(2c + d) \left[\frac{1}{a} + \frac{1}{b} + c \left(\frac{1}{a} + \frac{1}{b} \right)^2 - \frac{d}{ab} \right]. \quad (5)$$

The spin trace averages over two electron spins and two incoming photon polarizations, a factor of one quarter. It follows by expanding both diagrams and applying the even gamma-matrix trace identities. The code checks the polynomial identity, then evaluates the same amplitude using independent complex Dirac matrices and explicit physical polarizations.

3 Laboratory cross section and threshold

$$s = m^2 + 2mE_\gamma, \quad b = M^2 - 2mE_X, \quad E_{\gamma,\text{th}} = M + \frac{M^2}{2m}, \quad (6)$$

$$\lambda(s, m^2, M^2) = [s - (m + M)^2][s - (m - M)^2], \quad (7)$$

$$E_X^\pm = \frac{(E_\gamma + m)(s + M^2 - m^2) \pm E_\gamma \sqrt{\lambda}}{2s}, \quad (8)$$

$$\frac{d\sigma_{\gamma e \rightarrow Xe}}{dE_X} = \frac{\epsilon^2 e^4 (\hbar c)^2}{32\pi m E_\gamma^2} F, \quad E_X^- \leq E_X \leq E_X^+. \quad (9)$$

Outside the endpoints or below threshold the cross section is zero. Integrating the expression numerically agrees with the derived analytic primitive and Reference 15. At zero mass the integrated result is Klein–Nishina times the squared mixing. These are free-electron, tree-level statements; they do not establish that Compton scattering equals total material attenuation.

$$\Phi_X^{\text{Park}}(x) = \int_{\max(1 \text{ MeV}, E_{\text{th}})}^{40 \text{ MeV}} \frac{\Phi_\gamma(E)}{\sigma_{\text{KN}}(E)} \frac{d\sigma_{\text{prod}}(E, x)}{dx} dE, \quad \Phi_\gamma(E) = 0.58 \times 10^{18} \frac{P}{\text{MW}} e^{-E/(0.91 \text{ MeV})}. \quad (10)$$

The kernel enforces the additional angular support. Figure 1 uses 1 GW and the three original masses. Open points are vector coordinates extracted from the supplied Park PDF in v2, reused as source data. The original calculation is not reproduced in full; changing the source cutoff or fitting an overall factor is not a verified remedy. The gamma spectrum is an empirical fit above 0.2 MeV, while Park’s stated integration cutoff is 1 MeV.

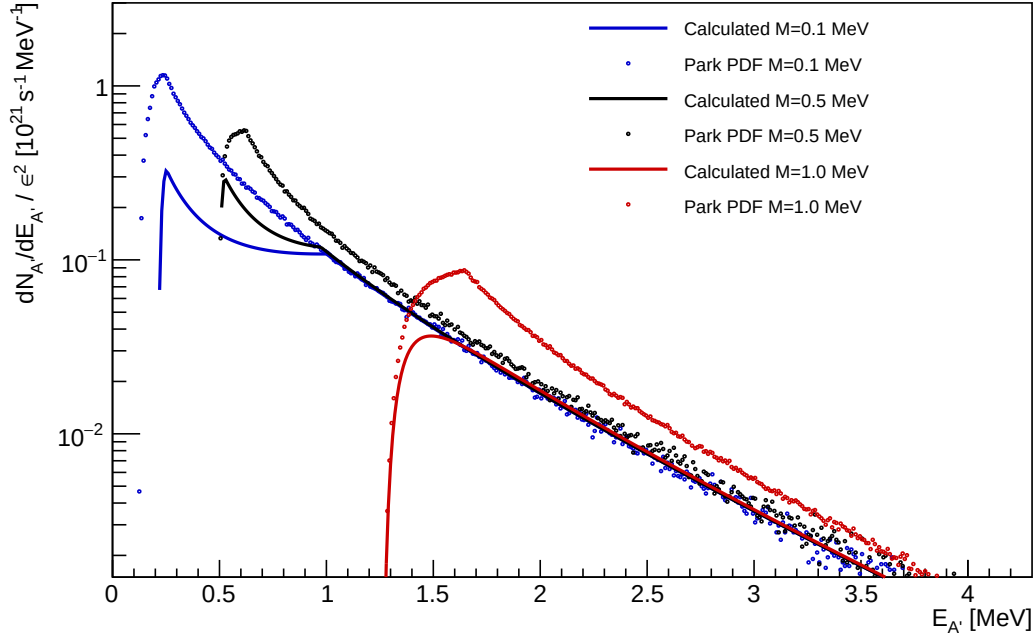


Figure 1: Recalculation versus Park Figure 1. Solid curves are computed; markers are extracted source data.

4 Detection by the inverse process

Time reversal gives the same invariant squared amplitude. For an unpolarized incident massive vector the average over three polarizations adds the factor two thirds relative to production. For incident energy E , momentum p_X , and outgoing photon energy w ,

$$p_X = \sqrt{E^2 - M^2}, \quad s = m^2 + M^2 + 2mE, \quad a = M^2 + 2mE, \quad b = -2mw, \quad (11)$$

$$w_{\pm} = \frac{(s - m^2)(E + m \pm p_X)}{2s}, \quad (12)$$

$$\frac{d\sigma_{Xe \rightarrow \gamma e}^{\text{unpol}}}{dw} = \frac{2}{3} \frac{\epsilon^2 e^4 (\hbar c)^2 F}{32\pi m p_X^2}. \quad (13)$$

The inverse massless limit is two thirds of Klein–Nishina. This is appropriate for the Park convention. It is not the factor used for oscillation-produced transverse states. Full containment gives $E_{\text{vis}} = E$ after combining electron and photon energy. A real detector needs a response function; replacing it by full containment is an explicit approximation.

5 Medium production and conversion

$$m_\gamma^2 = 4\pi\alpha n_e/m, \quad (14)$$

$$(\Delta m^2)^2 = (M^2 - m_\gamma^2)^2 + 2\epsilon^2 M^2 (M^2 + m_\gamma^2), \quad (15)$$

$$P_{\gamma \rightarrow X} = \frac{\epsilon^2 M^4}{(\Delta m_R^2)^2 + E^2 \Gamma_R^2}, \quad \Gamma_R = \frac{\hbar c}{L_{\text{abs}}}, \quad (16)$$

$$P_{X \rightarrow \gamma} = \frac{\epsilon^2 M^4}{(\Delta m_D^2)^2}, \quad \Phi_X^{\text{med}}(E) = \Phi_\gamma(E) P_{\gamma \rightarrow X}(E). \quad (17)$$

These are Danilov Eqs. (5), (6), and (9), not all-orders exact probabilities. The earlier implementation incorrectly used 4 instead of 2 in the splitting and treated the detector expression as valid at resonance. Equation (9) explicitly neglects absorption away from resonance. We omit a conservative ten-percent mass window around either plasma mass. A finite-size, absorbing, inhomogeneous detector calculation is needed within it. We set both plasma masses to 20 eV and the reactor absorption length to 10 cm, following the paper’s illustrative choices. At masses well above the plasma scale both probabilities approach the squared mixing. At much smaller masses their product scales as $\epsilon^4 (M/m_\gamma)^8$ when damping is negligible. The direct oscillation source preserves photon energy; the Compton-convolved Park source redistributes it. They are alternative source prescriptions and are not multiplied together.

6 Event rate, statistics, and comparisons

$$N_e = M_{\text{det}}[\text{kg}] 10^3 N_A Z/A, \quad (18)$$

$$N_{\text{med}} = \frac{N_e T \eta}{4\pi R^2} \int_{3 \text{ MeV}}^{8 \text{ MeV}} \Phi_\gamma(E) P_{\gamma \rightarrow X}(E) P_{X \rightarrow \gamma}(E) \sigma_{\text{KN}}(E) dE, \quad (19)$$

$$N_{\text{med}}(\epsilon_{95}, M) = 195.7, \quad 0 < \epsilon_{95} \leq 0.01. \quad (20)$$

Inputs are 2.9 GW, 28 m, 187 kg of CsI, 160 live days, $Z = 108$ and $A = 259.8099219$ g/mol. The measured excess quoted by Danilov is 30.7 ± 100.6 events. Their one-sided cap is 195.7 selected

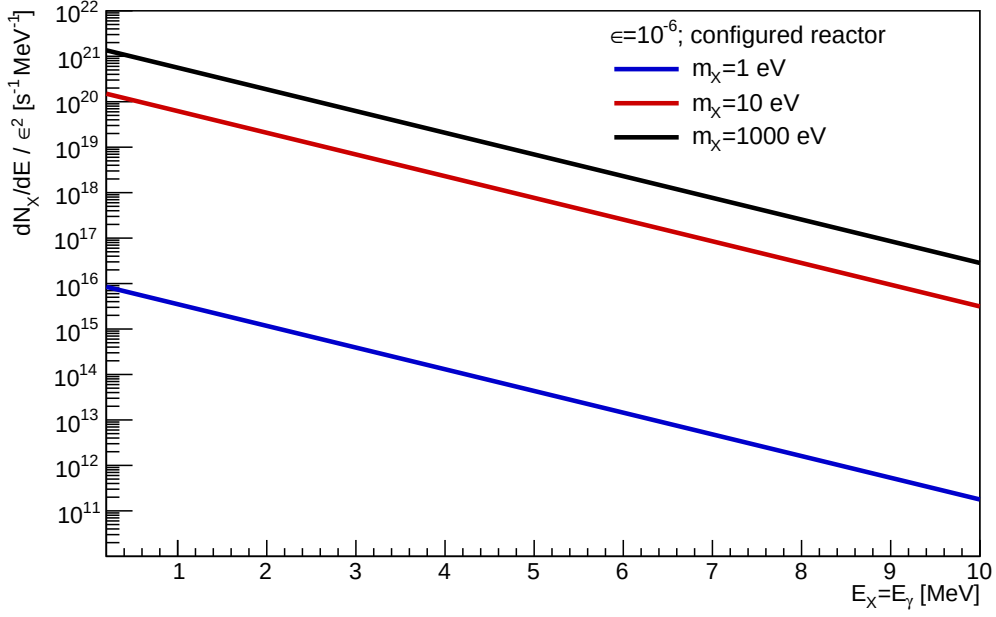


Figure 2: Medium source at the configured reactor power. The normalization is evaluated at $\epsilon = 10^{-6}$ and divided by its square.

events. Their assumed anti-Compton suppression corresponds to $\eta = 1/6$, approximately 1174 preselection events. It is an estimate, not a measured dark-photon efficiency. We also show unit efficiency to expose its effect. The earlier report used the selected-event cap with unit efficiency only and therefore omitted this correction. The Park-convention recalculation uses the convolved source, massive inverse cross section, unit efficiency and 1.96(100.6) events. Park labels this 95% CL; for a centered Gaussian it corresponds to a 97.5% one-sided cap. It must not be silently equated with the Danilov one-sided convention. The red line is Park's quoted 2.1×10^{-5} bound, not a newly digitized data set. The black line is extracted directly from Danilov's vector PDF. The independent medium calculation is not tuned to either line. Differences remain because the published normalization, source prescription and detector acceptance are not fully reproduced.

7 Validity and actual verification

The production kernel has exact two-body massive kinematics within tree-level free-electron QED. The medium detector model uses relativistic transverse modes, $M \ll m_e, E$, and is reported only up to 10 keV for a 3–8 MeV ROI. Plasma masses and absorption are constant inputs. Material attenuation, atomic binding, rescattering, escape of photons, finite geometry, and energy-dependent selection are not simulated. Thus the medium limit is conditional on this response and source model; it is not a validated detector-level TEXONO exclusion. The executed checks include independent matrix traces and Ward identities, the Reference 15 total, analytic integration, both massless scattering limits, the corrected splitting coefficient, root closure, doubled quadrature resolution, and rejection of invalid domains. Every plotted curve is checked for finite in-frame points and its presence in the canvas at export. The exported images are inspected visually. The original full Park spectrum and Danilov numerical curve are not reproduced by the independent prediction; their differences are visible.

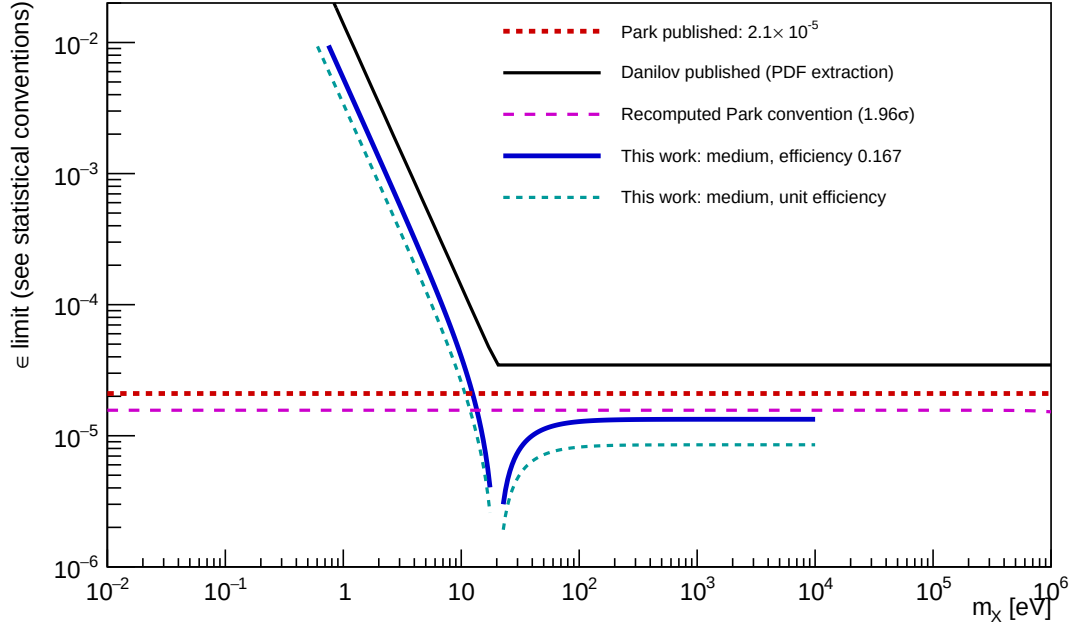


Figure 3: TEXONO-only comparison. Gaps near 20 eV mark the unclaimed resonance region. Medium curves end at 10 keV; the exact massive Park-convention calculation extends to 1 MeV.

References

- [1] H. K. Park, Phys. Rev. Lett. 119, 081801 (2017), local source papers/darkphoton_v3/park2017.pdf, Eqs. (1)–(7), Figs. 1–2.
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- [3] P. Gondolo and G. Raffelt, Phys. Rev. D 79, 107301 (2009), vector appendix, local source papers/darkphoton_v3/reference15.pdf.